

	clustered novelty	diffuse shift
off-manifold	everything works; anchors interpret. Best pipelines $f^*=0.008$ – 0.020 (Tables 5, 23)	detected only by the mean-shift monitor on the <i>frozen trunk</i> ($f^*=0.006$ – 0.010); head spaces nearly blind (Tables 28, 29)
on-manifold	SparKer’s local kernels detect the clump ($f^*=0.012$ – 0.039) but distance ranking is below chance, pools starve, Mahalanobis censors: detection without anything to anchor (Table 30)	the doubly-hard corner, covered by both failure modes

Table 1: Where discovery works, where only detection works, and where the method abstains: the failure 2×2 , every cell measured in the $f^*(2\sigma)$ currency. Locality (SparKer) detects clumps; mean-shift (Mahalanobis) detects smears; anchors interpret only what is both clustered and outlying — and the pipeline reports when it cannot. *Coverage:* Cells summarise Tables 5–30; the derived label-free cut abstains wherever estimated novel mass is below clusterability, which governs the boundary between cells.

Sigma-currency tables (exps 146/148), CIFAR-10 leakage-free lineage — generated September 8, 2026

$f^*(2\sigma)$ = smallest injected fraction at which the median expected significance of a dataset-level test reaches 2σ (200 null / 50 signal toys, $N_D=5000$); lower is better. Post tables: bold = best test per arm; † = the derived cut declined at some fraction below the crossing (the space is the frozen one there); purity ‘–’ = declined at $f=0.03$; ‘(n/6)’ = arm still running.

space	probe	per-ev.	$f^*(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.930	0.01	0.057	0.020	0.018	0.015–0.021
→ res (residual)	0.923	0.23	0.023	0.033	0.022	0.012–0.039
→ res (concat)	0.957	0.12	0.024	0.018	0.019	0.016–0.021
→ res-nplm (residual)	0.926	0.22	0.023	0.029	0.017	0.015–0.019
→ res-nplm (concat)	0.946	0.10	0.025	0.020	0.019	0.016–0.021
SupCon+SIGReg ($\lambda=5$)	0.912	0.02	$> 0.1^4$	0.031	0.037	0.032–0.042
→ res (residual)	0.902	0.01	$> 0.1^3$	0.044	0.069	0.044–0.091
→ res (concat)	0.949	0.01	$> 0.1^3$	0.035	0.042	0.027–0.060
→ res-nplm (residual)	0.921	0.16	0.044	0.038	0.022	0.017–0.025
→ res-nplm (concat)	0.943	0.06	$> 0.1^2$	0.033	0.030	0.020–0.036
NPLM-dist. (sup.)	0.788	0.17	0.038	0.039	0.027	0.024–0.040
→ res (residual)	0.884	0.03	$> 0.1^2$	0.046	0.046	0.044–0.077
→ res (concat)	0.908	0.03	$> 0.1^2$	0.042	0.043	0.033–0.059
→ res-nplm (residual)	0.841	0.01	$> 0.1^2$	0.044	0.039	0.024–0.049
→ res-nplm (concat)	0.879	0.09	0.047 ¹	0.045	0.031	0.018–0.043
NPLM-dist.+classwise SIGReg	0.938	0.21	0.032	0.033	0.027	0.024–0.029
→ res (residual)	0.917	0.03	$> 0.1^2$	0.037	0.048	0.039–0.084
→ res (concat)	0.958	0.03	0.034 ¹	0.035	0.047	0.037–0.078
→ res-nplm (residual)	0.948	0.07	0.034	0.036	0.043	0.028–0.079
→ res-nplm (concat)	0.966	0.17	0.023	0.035	0.037	0.018–0.046
SimCLR	0.808	0.00	$> 0.1^1$	0.027	0.056	0.056–0.056
VISReg/LeJEPa	0.832	0.01	$> 0.1^1$	0.047	0.091	0.091–0.091
NPLM (unsup.)	0.799	0.05	$> 0.1^1$	0.086	$> 0.1^1$	–
SupCon+SIGReg (repulse)	0.871	0.12	0.047	0.039	0.026	0.026–0.026

Table 2: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. f^* is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-10, leakage-free lineage, single holdout; draws 4, 7, 8, 9; 20/24 spaces on every draw; 200 null / 50 signal toys per fraction, $N_D=5000$; median over draws, superscript = draws where the test never reached 2σ by $f=0.1$.

space	probe	per-ev.	$f^*(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.955	0.03	$> 0.1^2$	0.024	0.023	0.010–0.036
→ res (residual)	0.925	0.05	$> 0.1^3$	0.025	0.028	0.023–0.030
→ res (concat)	0.963	0.05	0.057 ¹	0.024	0.018	0.017–0.026
→ res-nplm (residual)	0.929	0.10	$> 0.1^3$	0.025	0.027	0.016–0.048
→ res-nplm (concat)	0.963	0.07	$> 0.1^3$	0.023	0.022	0.010–0.030
SupCon+SIGReg ($\lambda=5$)	0.947	0.05	0.055	0.022	0.023	0.014–0.026
→ res (residual)	0.906	0.07	$> 0.1^3$	0.028	0.027	0.025–0.032
→ res (concat)	0.959	0.05	0.056	0.020	0.019	0.016–0.026
→ res-nplm (residual)	0.932	0.05	0.055 ¹	0.022	0.024	0.016–0.027
→ res-nplm (concat)	0.941	0.05	0.059	0.021	0.023	0.014–0.026
NPLM-dist. (sup.)	0.857	0.07	$> 0.1^2$	0.025	0.034	0.017–0.046
→ res (residual)	0.892	0.11	$> 0.1^2$	0.026	0.026	0.025–0.033
→ res (concat)	0.917	0.10	$> 0.1^3$	0.026	0.028	0.026–0.041
→ res-nplm (residual)	0.905	0.08	$> 0.1^2$	0.032	0.026	0.021–0.031
→ res-nplm (concat)	0.924	0.07	$> 0.1^2$	0.028	0.025	0.022–0.029
NPLM-dist.+classwise SIGReg	0.944	0.12	$> 0.1^2$	0.021	0.025	0.015–0.029
→ res (residual)	0.911	0.06	$> 0.1^3$	0.025	0.027	0.024–0.030
→ res (concat)	0.956	0.06	$> 0.1^2$	0.025	0.027	0.023–0.032
→ res-nplm (residual)	0.878	0.04	$> 0.1^3$	0.027	0.027	0.026–0.030
→ res-nplm (concat)	0.955	0.04	$> 0.1^2$	0.022	0.026	0.018–0.030
SimCLR	0.799	0.01	$> 0.1^1$	0.046	0.052	0.052–0.052
VISReg/LeJEPa	0.797	0.03	$> 0.1^1$	0.059	0.047	0.047–0.047
NPLM (unsup.)	0.757	0.01	$> 0.1^1$	0.040	0.031	0.031–0.031
SupCon+SIGReg (repulse)	0.823	0.00	$> 0.1^1$	0.031	0.029	0.029–0.029

Table 3: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. f^* is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-100, leakage-free lineage, single holdout; draws 4, 43, 48, 57; 20/24 spaces on every draw; 200 null / 50 signal toys per fraction, $N_D=5000$; median over draws, superscript = draws where the test never reached 2σ by $f=0.1$.

arm	draw	frozen SparKer	dist/95th	dist/derived	np/derived
SupCon	h4	0.015	0.027 (MMD)	0.014 (MMD) [†]	0.014 (MMD) [†]
	h7	0.021	0.035 (SparKer (anch.))	0.033 (SparKer (anch.))	0.018 (SparKer)
	h8	0.018	0.016 (Eucl. (anch.))	0.017 (MMD)	0.011 (Eucl. (anch.))
	h9	0.018	0.008 (Eucl.)	0.011 (Eucl. (anch.))	0.009 (SparKer)
SupCon+SIGReg ($\lambda=5$)	h4	0.042	0.021 (Eucl. (anch.))	0.028 (MMD)	0.028 (Eucl.)
	h7	0.034	0.021 (Eucl. (anch.))	0.036 (SparKer) [†]	0.036 (SparKer) [†]
	h8	0.040	0.015 (MMD)	0.021 (MMD)	0.020 (Eucl. (anch.))
	h9	0.032	0.016 (Eucl. (anch.))	0.026 (MMD) [†]	0.026 (MMD) [†]
NPLM-dist. (sup.)	h4	0.040	0.024 (Eucl. (anch.))	0.037 (Eucl. (anch.))	0.033 (Eucl. (anch.))
	h7	0.029	0.022 (Eucl. (anch.))	0.039 (Eucl. (anch.))	0.021 (Eucl. (anch.))
	h8	0.024	0.016 (SparKer (anch.))	0.015 (Eucl. (anch.))	0.016 (Eucl. (anch.))
	h9	0.024	0.009 (SparKer (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
NPLM-dist.+classwise SIGReg	h4	0.025	0.019 (Eucl. (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
	h7	0.029	0.018 (Eucl. (anch.))	0.027 (SparKer (anch.))	0.011 (Eucl. (anch.))
	h8	0.024	0.015 (Eucl. (anch.))	0.012 (Eucl. (anch.))	0.009 (Eucl. (anch.))
	h9	0.028	0.016 (Maha.)	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))

Table 4: Does the discovery loop beat the frozen space? Frozen-space SparKer $f^*(2\sigma)$ against the best post-discovery test (named in parentheses) under three loop variants: the campaign’s distance pool with the 95th-percentile cut, the distance pool with the derived cut, and the density-ratio pool with the derived cut (the paper’s construction). Bold = the loop wins on that draw. *Coverage:* CIFAR-10, four parents x draws 4, 7, 8, 9; 48 complete (arm, draw, variant) cells, the loop beats the frozen space in 38; [†] = the derived cut had declined at the crossing fraction, so that number is the frozen space’s and is not counted as a win; ‘-’ = variant not run or still running.

dataset / backbone	best frozen (space / test, median f^*)		best discovery (arm / test)		gain
CIFAR-100 (scratch, 4 draws)	SupCon→res (concat) / SparKer	0.018	SupCon→res-nplm / SparKer ^e	0.008	2.2×
CIFAR-10 (scratch, 4 draws)	SupCon→res-nplm (resid.) / SparKer	0.017	SupCon / Eucl. (anch.)	0.013	1.3×
Galaxy10 / lejepa	SupCon / MMD	0.023	SupCon→res / Eucl. (anch.)	0.013	1.7×
Galaxy10 / visreg	SupCon→res (concat) / MMD	0.026	SupCon→res / Eucl. (anch.)	0.015	1.7×
Galaxy10 / dino	SupCon+SIGReg / MMD	0.029	SupCon / Eucl. (anch.)	0.019	1.5×
transductive ref. (lejepa)	gcd-ft / Maha.	0.013	–	–	sees the class
transductive ref. (visreg)	gcd-ft / Maha.	0.015	–	–	sees the class
transductive ref. (dino)	gcd-ft / MMD	0.028	–	–	sees the class

Table 5: The chosen algorithm, per dataset: best frozen space vs. best discovery pipeline in the same currency. Discovery wins everywhere, always through an anchor-aware test; the transductive GCD rows are the ceiling given the whole novel class unlabelled. *Coverage:* Medians over draws (censored cells imputed at 0.15); CIFAR-10 discovery = np pool + derived cut ($n_{\min}=10$), Galaxy10 discovery = np pool + derived cut ($n_{\min}=5$), both skip-on-refuse; frozen candidates exclude the transductive GCD arms (shown separately). Declined-cut crossings are excluded from the discovery medians; ^e = CIFAR-100 medians cover the engaged draws only (the cut abstains on 33/48 cells at $b=0.01$).

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	0.00	0.063	> 0.1	0.072	0.027	0.036	0.029
→ res (residual)	0.039	0.04	0.074	0.026	0.039	0.043	0.047	0.026
→ res-nplm (residual)	0.016	0.08	0.082	0.029	0.065	0.045	0.037	0.031
SupCon+SIGReg ($\lambda=5$)	0.042	0.01	0.033	0.021	> 0.1	0.027	0.031	0.039
→ res (residual)	0.091	0.01	0.068	0.043	0.035	0.036	0.044	0.045
→ res-nplm (residual)	0.023	0.10	0.027	0.017	0.077	0.028	0.029	0.037
NPLM-dist. (sup.)	0.040	0.10	> 0.1	0.024	0.087	0.046	0.046	0.043
→ res (residual)	0.077	0.01	> 0.1	> 0.1	0.081	0.035	0.043	0.041
→ res-nplm (residual)	0.049	0.01	> 0.1	0.047	0.087	0.062	0.083	0.044
NPLM-dist.+classwise SIGReg	0.025	0.20	0.034	0.019	0.033	0.044	0.028	0.024
→ res (residual)	0.084	0.01	0.027	0.027	0.033	0.033	0.029	0.027
→ res-nplm (residual)	0.079	0.03	0.043	0.022	0.037	0.036	0.047	0.041

Table 6: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, 95th-pct. cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 †	0.017†	–
→ res (residual)	0.039	0.03	0.042	0.039	0.068	0.039	0.040	0.032
→ res-nplm (residual)	0.016	0.01	0.047	0.039	0.049	0.034	0.027	0.024
SupCon+SIGReg ($\lambda=5$)	0.042	0.00	0.038	0.062	> 0.1	0.028	0.047	0.044
→ res (residual)	0.091	0.00	> 0.1	> 0.1	0.078	0.038	0.049	0.087
→ res-nplm (residual)	0.023	0.07	0.072	0.024	> 0.1	0.039	0.040	0.026
NPLM-dist. (sup.)	0.040	–	0.077	0.037	0.072	0.045	0.043	0.043
→ res (residual)	0.077	0.02	> 0.1	> 0.1	> 0.1	0.038	0.052	0.060
→ res-nplm (residual)	0.049	–	0.089	0.065	0.089	0.080	0.049 †	0.074
NPLM-dist.+classwise SIGReg	0.025	–	0.028†	0.020	0.074	0.029†	0.028†	0.021
→ res (residual)	0.084	–	> 0.1	0.050	0.083	0.041	0.047	0.050
→ res-nplm (residual)	0.079	–	0.049	0.050	0.046	0.043	0.067	0.050

Table 7: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 [†]	0.017 [†]	–
→ res (residual)	0.039	0.21	0.033	0.015	0.024	0.041	0.033	0.029
→ res-nplm (residual)	0.016	0.14	0.035	0.020	0.039	0.041	0.037	0.034
SupCon+SIGReg ($\lambda=5$)	0.042	0.13	0.028	0.030	0.083	0.034	0.042	0.030
→ res (residual)	0.091	0.15	0.082	0.030	0.033	0.042	0.060	0.047
→ res-nplm (residual)	0.023	0.12	0.044	0.018	0.074	0.037	0.041	0.021
NPLM-dist. (sup.)	0.040	–	0.087	0.033	0.084	0.049	0.073	0.042
→ res (residual)	0.077	0.43	0.056	0.020	0.029	0.046	0.046	0.034
→ res-nplm (residual)	0.049	–	0.089	0.046	0.089	0.080	0.049 [†]	0.062
NPLM-dist.+classwise SIGReg	0.025	–	0.028 [†]	0.020	0.072	0.029 [†]	0.027 [†]	0.036
→ res (residual)	0.084	–	0.079	0.050	0.046	0.044	0.045	0.050
→ res-nplm (residual)	0.079	–	0.044	0.050	0.046	0.043	0.057	0.050

Table 8: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	> 0.1	> 0.1	0.043	0.047	0.052	0.035
SupCon+SIGReg ($\lambda=5$)	0.034	0.03	0.042	0.021	> 0.1	0.056	0.040	0.043
NPLM-dist. (sup.)	0.029	0.08	0.082	0.022	0.038	0.044	0.029	0.023
NPLM-dist.+classwise SIGReg	0.029	0.17	0.029	0.018	0.024	0.037	0.028	0.025

Table 9: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	0.071	> 0.1	0.036	0.056	0.040	0.033
SupCon+SIGReg ($\lambda=5$)	0.034	-	0.079	-	0.090	0.079	0.036 †	-
NPLM-dist. (sup.)	0.029	0.05	> 0.1	0.039	0.044	0.042	0.042	0.041
NPLM-dist.+classwise SIGReg	0.029	0.10	0.042	0.029	0.035	0.052	0.040	0.027

Table 10: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.68	0.022	0.020	0.021	0.052	0.018	0.020
→ res (residual)	0.026	0.63	0.017	0.014	0.016	0.049	0.018	0.017
→ res-nplm (residual)	0.019	0.61	0.021	0.021	0.021	0.065	0.022	0.023
SupCon+SIGReg ($\lambda=5$)	0.034	-	0.079	-	0.090	0.072	0.036 †	-
→ res (residual)	0.089	0.12	0.059	0.030	0.027	0.079	0.066	0.046
→ res-nplm (residual)	0.025	0.63	0.039	0.016	0.038	0.044	0.028	0.018
NPLM-dist. (sup.)	0.029	0.18	0.044	0.021	0.033	0.043	0.041	0.029
→ res (residual)	0.045	-	0.044	0.050	0.031	0.076	0.047	0.050
→ res-nplm (residual)	0.049	-	0.092	-	0.089	0.073	0.072	-
NPLM-dist.+classwise SIGReg	0.029	0.67	0.043	0.011	0.018	0.049	0.033	0.019
→ res (residual)	0.039	0.22	0.036	0.030	0.023	0.068	0.029	0.030
→ res-nplm (residual)	0.052	-	0.068	0.050	0.026 †	0.072	0.048	0.050

Table 11: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.039	0.016	0.023	0.018	0.021	0.017
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.026	0.019	> 0.1	0.015	0.026	0.023
NPLM-dist. (sup.)	0.024	0.16	0.037	0.017	0.019	0.019	0.021	0.016
NPLM-dist.+classwise SIGReg	0.024	0.18	0.017	0.015	0.022	0.022	0.017	0.015

Table 12: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.052	> 0.1	0.038	0.017	0.031	0.027
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.024	0.076	0.062	0.021	0.029	0.029
NPLM-dist. (sup.)	0.024	0.59	0.063	0.015	0.025	0.023	0.023	0.017
NPLM-dist.+classwise SIGReg	0.024	0.55	0.042	0.012	0.039	0.036	0.035	0.021

Table 13: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.45	0.014	0.011	0.015	0.019	0.017	0.014
→ res (residual)	0.018	0.42	0.017	0.020	0.016	0.019	0.016	0.020
→ res-nplm (residual)	0.018	0.56	0.021	0.015	0.012	0.017	0.016	0.015
SupCon+SIGReg ($\lambda=5$)	0.040	0.22	0.024	0.020	0.024	0.025	0.032	0.035
→ res (residual)	0.044	-	0.046	0.050	0.045	0.021 †	0.041	0.050
→ res-nplm (residual)	0.021	0.48	0.038	0.020	0.025	0.021	0.024	0.036
NPLM-dist. (sup.)	0.024	0.64	0.046	0.016	0.041	0.024	0.018	0.017
→ res (residual)	0.046	0.30	0.028	0.030	0.028	0.022	0.028	0.030
→ res-nplm (residual)	0.029	0.18	0.027	0.020	0.026	0.027	0.027	0.028
NPLM-dist.+classwise SIGReg	0.024	0.73	0.075	0.009	0.037	0.044	0.040	0.021
→ res (residual)	0.054	0.40	0.026	0.011	0.019	0.019	0.025	0.024
→ res-nplm (residual)	0.033	0.36	0.074	0.020	0.027	0.022	0.037	0.020

Table 14: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.02	0.008	0.010	0.017	0.018	0.010	0.009
SupCon+SIGReg ($\lambda=5$)	0.032	0.01	0.025	0.016	0.040	0.040	0.025	0.025
NPLM-dist. (sup.)	0.024	0.14	0.017	0.014	0.015	0.018	0.014	0.009
NPLM-dist.+classwise SIGReg	0.028	0.17	0.016	0.022	0.016	0.029	0.017	0.017

Table 15: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.013	0.011	0.025	0.020	0.014	0.018
SupCon+SIGReg ($\lambda=5$)	0.032	–	0.047	> 0.1	0.048	0.026 †	0.027†	0.050
NPLM-dist. (sup.)	0.024	0.82	0.036	0.020	0.024	0.029	0.021	0.027
NPLM-dist.+classwise SIGReg	0.028	0.51	0.026	0.020	0.028	0.036	0.026	0.020

Table 16: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.28	0.010	0.013	0.016	0.025	0.009	0.010
→ res (residual)	0.012	0.50	0.009	0.010	0.009	0.019	0.009	0.010
→ res-nplm (residual)	0.015	0.75	0.014	0.013	0.014	0.016	0.009	0.010
SupCon+SIGReg ($\lambda=5$)	0.032	–	0.047	0.050	0.045	0.026 †	0.027†	0.050
→ res (residual)	0.049	0.15	0.035	0.030	0.027	0.037	0.028	0.030
→ res-nplm (residual)	0.017	0.59	0.024	0.010	0.018	0.028	0.017	0.016
NPLM-dist. (sup.)	0.024	0.84	0.027	0.020	0.023	0.029	0.023	0.020
→ res (residual)	0.044	0.16	0.028	0.030	0.027	0.034	0.028	0.030
→ res-nplm (residual)	0.024	0.55	0.082	0.016	0.027	0.027	0.025	0.026
NPLM-dist.+classwise SIGReg	0.028	0.86	0.044	0.020	0.025	0.038	0.025	0.020
→ res (residual)	0.042	–	0.044	0.050	0.028 †	0.029†	0.036	0.050
→ res-nplm (residual)	0.028	0.20	0.075	0.030	0.025	0.044	0.025	0.030

Table 17: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.036	0.07	> 0.1	0.023	> 0.1	0.018	0.028	0.024
→ res (residual)	0.030	–	0.063 [†]	–	> 0.1	0.042 [†]	0.030 [†]	–
→ res-nplm (residual)	0.048	–	> 0.1	–	> 0.1	0.028 [†]	0.052 [†]	–
SupCon+SIGReg ($\lambda=5$)	0.026	–	0.059 [†]	–	0.056 [†]	0.016 [†]	0.026 [†]	–
→ res (residual)	0.032	–	0.052 [†]	–	> 0.1	0.047 [†]	0.037 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.024 [†]	0.026 [†]	–
NPLM-dist. (sup.)	0.041	–	> 0.1	–	> 0.1	0.019 [†]	0.048 [†]	–
→ res (residual)	0.033	–	> 0.1	–	> 0.1	0.047 [†]	0.034 [†]	–
→ res-nplm (residual)	0.031	–	> 0.1	–	> 0.1	0.047 [†]	0.033 [†]	–
NPLM-dist.+classwise SIGReg	0.029	0.04	> 0.1	0.034	> 0.1	0.037	0.033	0.025
→ res (residual)	0.030	–	> 0.1	–	> 0.1	0.036 [†]	0.037 [†]	–
→ res-nplm (residual)	0.030	–	> 0.1	–	> 0.1	0.041 [†]	0.030 [†]	–

Table 18: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.023	–	> 0.1	–	0.062 [†]	0.027 [†]	0.023 [†]	–
→ res (residual)	0.023	–	0.028 [†]	–	0.047 [†]	0.033 [†]	0.023 [†]	–
→ res-nplm (residual)	0.026	–	> 0.1	–	> 0.1	0.025 [†]	0.028 [†]	–
SupCon+SIGReg ($\lambda=5$)	0.022	–	0.082 [†]	–	0.059 [†]	0.027 [†]	0.021 [†]	–
→ res (residual)	0.026	–	0.027 [†]	–	0.040 [†]	0.029 [†]	0.026 [†]	–
→ res-nplm (residual)	0.024	–	> 0.1	–	> 0.1	0.028 [†]	0.023 [†]	–
NPLM-dist. (sup.)	0.046	–	> 0.1	–	> 0.1	0.046 [†]	0.062 [†]	–
→ res (residual)	0.026	–	0.019 [†]	–	0.038 [†]	0.025 [†]	0.026 [†]	–
→ res-nplm (residual)	0.026	–	0.084 [†]	–	> 0.1	0.033 [†]	0.027 [†]	–
NPLM-dist.+classwise SIGReg	0.027	–	> 0.1	–	> 0.1	0.033 [†]	0.029 [†]	–
→ res (residual)	0.024	–	0.027 [†]	–	> 0.1	0.025 [†]	0.025 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.031 [†]	0.028 [†]	–

Table 19: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 43, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.010	–	0.045 [†]	–	> 0.1	0.026 [†]	0.013[†]	–
→ res (residual)	0.028	–	> 0.1	–	> 0.1	0.016[†]	0.029 [†]	–
→ res-nplm (residual)	0.016	0.02	0.015	0.022	0.035	0.013	0.008	0.016
SupCon+SIGReg ($\lambda=5$)	0.014	0.12	0.024	0.011	0.044	0.017	0.016	0.016
→ res (residual)	0.028	–	> 0.1	–	> 0.1	0.027[†]	0.030 [†]	–
→ res-nplm (residual)	0.016	0.08	0.025	0.014	0.038	0.016	0.014	0.015
NPLM-dist. (sup.)	0.017	0.35	0.033	0.024	0.029	0.016	0.026	0.010
→ res (residual)	0.025	–	0.036 [†]	–	0.031 [†]	0.017[†]	0.023 [†]	–
→ res-nplm (residual)	0.021	–	0.048 [†]	–	0.025 [†]	0.015[†]	0.023 [†]	–
NPLM-dist.+classwise SIGReg	0.015	0.40	0.026	0.014	0.025	0.016	0.015	0.008
→ res (residual)	0.025	–	0.077 [†]	–	0.039 [†]	0.018[†]	0.025 [†]	–
→ res-nplm (residual)	0.026	–	> 0.1	–	> 0.1	0.023[†]	0.026 [†]	–

Table 20: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 48, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.024	–	> 0.1	–	0.039 [†]	0.036 [†]	0.027[†]	–
→ res (residual)	0.029	–	> 0.1	–	> 0.1	0.017[†]	0.028 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.028 [†]	0.026[†]	–
SupCon+SIGReg ($\lambda=5$)	0.024	–	> 0.1	–	0.059 [†]	0.021 [†]	0.019[†]	–
→ res (residual)	0.025	–	> 0.1	–	> 0.1	0.025[†]	0.028 [†]	–
→ res-nplm (residual)	0.024	–	0.052 [†]	–	0.033 [†]	0.017[†]	0.021 [†]	–
NPLM-dist. (sup.)	0.026	0.07	0.037 [†]	0.023	0.037 [†]	0.025	0.018	0.022
→ res (residual)	0.026	–	> 0.1	–	> 0.1	0.022[†]	0.027 [†]	–
→ res-nplm (residual)	0.025	–	0.076 [†]	–	0.052 [†]	0.024[†]	0.026 [†]	–
NPLM-dist.+classwise SIGReg	0.023	0.06	0.034	0.035	0.026	0.016	0.016	0.010
→ res (residual)	0.029	–	> 0.1	–	> 0.1	0.023[†]	0.034 [†]	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.022[†]	0.026 [†]	–

Table 21: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-100 holdout 57, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.892	0.05	.029	.052	.078	.037	>.1	.015	.048	.031	.022	.023	.015	.073	.066	.035	.042
SupCon+SIGReg ($\lambda=5$)	0.762	0.13	.023	>.1	>.1	.033	>.1	.015	.029	.029	.040	.037	.014	>.1	>.1	.028	.043
NPLM-dist. (sup.)	0.787	0.13	>.1	>.1	>.1	.021	>.1	.019	.063	.037	.026	.024	.029	.032	.087	.022	.031
SimCLR	0.829	0.05	.029	.046	>.1	.036	.070	.043	.039	.070	.045	.047	.078	.081	>.1	.095	>.1
SimCLR+SIGReg	0.836	0.06	.048	.065	>.1	.034	>.1	.035	.029	.039	.080	.062	.046	.095	>.1	.046	.093
NPLM-bilinear	0.808	0.08	.078	.042	>.1	.026	>.1	.038	.039	.056	.046	.052	.052	.070	.093	.049	.088
GCD (transductive)	0.899	0.30	.009	.013	.009	.031	.017	.019	.025	.029	.024	.019	.009	.017	.021	.031	.025
GCD+SIGReg (transd.)	0.872	0.48	.015	.019	.019	.043	.025	.014	.035	.040	.042	.029	.013	.027	.026	.030	.028
SupCon $\rightarrow$ res (residual)	0.910	0.11	.085	.048	>.1	.024	.025	.021	.040	.045	.043	.039	.036	.063	.072	.044	.040
SupCon $\rightarrow$ res (concat)	0.945	0.12	.084	.041	>.1	.024	.034	.019	.040	.042	.042	.038	.029	.056	.046	.042	.026
SupCon $\rightarrow$ res-nplm (residual)	0.887	0.08	>.1	.027	>.1	.033	>.1	.016	.041	.073	.045	.032	.029	.072	.089	.040	.078
SupCon $\rightarrow$ res-nplm (concat)	0.931	0.09	.070	.034	.070	.034	.052	.015	.042	.031	.026	.024	.023	.063	.073	.031	.041
SupCon+SIGReg $\rightarrow$ res (residual)	0.854	0.06	>.1	.043	>.1	.019	.041	.019	.036	.045	.040	.031	.043	.089	.081	.037	.072
SupCon+SIGReg $\rightarrow$ res (concat)	0.855	0.09	.031	.063	>.1	.021	.042	.017	.026	.037	.036	.031	.019	.092	.085	.028	.041
SupCon+classwise SIGReg ($\lambda=5$)	0.810	0.05	>.1	.043	>.1	.068	.070	.027	.036	.039	.031	.052	.066	.052	>.1	.047	.052

Table 22: Galaxy10 / lejepa: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / LeJEPa, 15 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.019	.044	.038	.023	.027	.027
Eucl. (anch.)	.010	.020	.018	.014	.020	.018
Maha.	.021	.026	.045	.015	.070	.026
MMD	.010	.071	.046	.019	.040	.040
SparKer	.010	.047	.044	.028	.026	.028
SparKer (anch.)	.010	.036	.042	.025	.025	.025
<i>SupCon</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.012	.013	.023	.010	.020	.013
Maha.	.033	.034	.059	.015	.025	.033
MMD	.026	.052	.043	.049	.059	.049
SparKer	.027	.066	.048	.043	.045	.045
SparKer (anch.)	.015	.074	.040	.044	.028	.040
<i>SupCon</i> $\rightarrow$ <i>res-nplm</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.064	.033	.097	.017	>.1	.064
Eucl. (anch.)	.010	.020	.036	.013	.020	.020
Maha.	.031	.027	>.1	.019	.096	.031
MMD	.021	.045	.069	.047	.037	.045
SparKer	.026	.042	.069	.036	.052	.042
SparKer (anch.)	.017	.027	.042	.041	.028	.028
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.038	.070	.079	.024	.069	.069
Eucl. (anch.)	.010	.020	.016	.014	.020	.016
Maha.	.018	.047	.072	.023	.056	.047
MMD	.018	.047	.042	.028	.041	.041
SparKer	.017	.062	.084	.019	.043	.043
SparKer (anch.)	.010	.052	.069	.018	.028	.028
<i>SupCon+SIGReg</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	.052	>.1	>.1	>.1	.150
Eucl. (anch.)	.015	.018	.025	.011	.020	.018
Maha.	.097	.039	>.1	.014	.029	.039
MMD	.026	.043	.041	.046	.045	.043
SparKer	.026	.088	.085	.041	.049	.049
SparKer (anch.)	.016	.056	.046	.029	.029	.029
<i>NPLM-dist.</i> (<i>sup.</i>) — engaged from d0: never, d3: never, d5: 0.05, d7: 0.1, d8: 0.02; median engaged purity 0.59						
Eucl.	>.1	.070 [†]	>.1	.021 [†]	.029 [†]	.070
Eucl. (anch.)	—	—	—	—	.020	.020
Maha.	>.1	>.1	>.1	.021 [†]	.045	.150
MMD	.016 [†]	.052 [†]	.037	.027 [†]	.027 [†]	.027
SparKer	.029 [†]	.028 [†]	.088 [†]	.023 [†]	.028 [†]	.028
SparKer (anch.)	—	—	—	—	.020	.020

Table 23: Galaxy10 / lejepa: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage*: Galaxy10 / lejepa, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 172 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.906	0.10	>.1	.065	>.1	.042	>.1	.019	.031	.034	.029	.019	.031	.047	.065	.021	.030
SupCon+SIGReg ($\lambda=5$)	0.823	0.09	>.1	.065	>.1	.029	>.1	.025	.018	.033	.037	.033	.036	.046	.076	.027	.043
NPLM-dist. (sup.)	0.738	0.13	>.1	.065	>.1	.019	.052	.024	.065	.043	.028	.029	.032	.040	.094	.017	.024
SimCLR	0.847	0.04	.075	.047	>.1	.031	>.1	.049	.044	.066	.052	.049	.072	.093	.090	.082	>.1
SimCLR+SIGReg	0.835	0.05	>.1	.090	>.1	.038	>.1	.040	.026	.044	.070	.046	.069	.091	>.1	.042	.092
NPLM-bilinear	0.672	0.06	>.1	.096	>.1	>.1	>.1	.044	.033	.034	.059	.034	.069	.063	.047	.097	>.1
GCD (transductive)	0.886	0.40	.015	.009	.012	.041	.018	.024	.021	.026	.029	.028	.010	.015	.017	.027	.019
GCD+SIGReg (transd.)	0.852	0.48	.013	.017	.026	.031	.026	.017	.024	.039	.033	.027	.022	.022	.032	.029	.019
SupCon $\rightarrow$ res (residual)	0.930	0.06	>.1	.037	>.1	.043	.037	.016	.025	.035	.036	.036	.026	.087	.082	.047	.043
SupCon $\rightarrow$ res (concat)	0.947	0.09	>.1	.040	.084	.031	.042	.019	.026	.034	.026	.025	.025	.045	.063	.028	.029
SupCon $\rightarrow$ res-nplm (residual)	0.890	0.11	>.1	.052	>.1	.024	.073	.019	.036	.033	.047	.027	.031	.065	.081	.022	.029
SupCon $\rightarrow$ res-nplm (concat)	0.923	0.12	>.1	.048	>.1	.033	.075	.018	.035	.033	.029	.024	.029	.049	.071	.023	.029
SupCon+SIGReg $\rightarrow$ res (residual)	0.867	0.05	>.1	.056	>.1	.017	.052	.025	.024	.039	.045	.038	.038	.086	.080	.043	.062
SupCon+SIGReg $\rightarrow$ res (concat)	0.888	0.09	>.1	.042	>.1	.019	.036	.021	.021	.033	.040	.029	.029	.043	.052	.030	.037
SupCon+classwise SIGReg ($\lambda=5$)	0.898	0.07	>.1	.048	>.1	.042	.070	.023	.031	.052	.042	.028	.041	.069	.089	.047	.059

Table 24: Galaxy10 / visreg: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / VISREG, 15 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.070	.038	.063	.024	.028	.038
Eucl. (anch.)	.015	.020	.021	.019	.020	.020
Maha.	>.1	.024	.056	.021	.042	.042
MMD	.017	.037	.040	.029	.027	.029
SparKer	.025	.056	.063	.024	.033	.033
SparKer (anch.)	.022	.042	.044	.021	.021	.022
<i>SupCon</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.015	.020	.024	.014	.014	.015
Maha.	.045	.042	.066	.019	.033	.042
MMD	.031	.026	.029	.026	.045	.029
SparKer	.025	.054	.084	.038	.039	.039
SparKer (anch.)	.013	.041	.077	.024	.029	.029
<i>SupCon</i> $\rightarrow$ <i>res-nplm</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	.034	.037	.052	.022	.025	.034
Eucl. (anch.)	.015	.021	.025	.013	.019	.019
Maha.	.056	.024	.049	.015	.028	.028
MMD	.024	.040	.039	.042	.038	.039
SparKer	.026	.048	.047	.024	.024	.026
SparKer (anch.)	.017	.025	.044	.020	.027	.025
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.077	.048	>.1	.016	>.1	.077
Eucl. (anch.)	.017	.022	.016	.013	.020	.017
Maha.	>.1	.026	.097	.023	.059	.059
MMD	.019	.036	.039	.041	.034	.036
SparKer	.034	.030	.062	.026	.045	.034
SparKer (anch.)	.018	.029	.040	.017	.025	.025
<i>SupCon+SIGReg</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.02, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.020	.018	.031	.010	.020	.020
Maha.	.063	.034	.072	.015	.027	.034
MMD	.028	.033	.043	.038	.029	.033
SparKer	.031	.052	.078	.044	.047	.047
SparKer (anch.)	.021	.034	.042	.037	.026	.034
<i>NPLM-dist.</i> (<i>sup.</i>) — engaged from d0: 0.02, d3: 0.05, d5: 0.1, d7: 0.03, d8: 0.1; median engaged purity 0.43						
Eucl.	.082 [†]	.041	>.1	.015 [†]	.026 [†]	.041
Eucl. (anch.)	—	—	—	.030	—	.030
Maha.	>.1	.039	>.1	.016 [†]	.044 [†]	.044
MMD	.018	.044	.044 [†]	.025	.037 [†]	.037
SparKer	.034 [†]	.032	.087	.017 [†]	.027 [†]	.032
SparKer (anch.)	—	—	—	.030	—	.030

Table 25: Galaxy10 / visreg: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage*: Galaxy10 / visreg, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 172 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.879	0.06	>.1	>.1	.063	.076	>.1	.031	.029	.025	.042	.037	.078	>.1	>.1	.049	.083
SupCon+SIGReg ($\lambda=5$)	0.804	0.04	>.1	.072	.069	>.1	>.1	.021	.025	.031	.047	.029	>.1	>.1	>.1	.079	>.1
NPLM-dist. (sup.)	0.775	0.10	>.1	>.1	>.1	.059	>.1	.037	.023	.029	.038	.037	.054	.086	.070	.038	.048
SimCLR	0.790	0.05	>.1	.066	>.1	>.1	>.1	.029	.039	.056	.067	.052	.070	.097	>.1	.066	>.1
SimCLR+SIGReg	0.801	0.04	>.1	.069	>.1	.076	>.1	.027	.029	.041	.088	.059	.069	.084	>.1	.059	.091
NPLM-bilinear	0.684	0.04	>.1	>.1	.082	>.1	>.1	.097	.042	.041	.093	>.1	.081	.089	.079	.083	.089
GCD (transductive)	0.821	0.07	>.1	.052	.037	.076	>.1	.019	.025	.031	.040	.028	.059	.087	.085	.069	.092
GCD+SIGReg (transd.)	0.831	0.12	.084	.066	.056	.073	>.1	.024	.026	.034	.039	.034	.052	>.1	.096	.066	.098
SupCon $\rightarrow$ res (residual)	0.891	0.05	>.1	.080	.062	.097	>.1	.023	.029	.047	.056	.033	.090	.098	.097	.081	.097
SupCon $\rightarrow$ res (concat)	0.903	0.06	>.1	>.1	.063	.074	>.1	.025	.025	.029	.048	.034	.074	>.1	.098	.073	.092
SupCon $\rightarrow$ res-nplm (residual)	0.862	0.07	>.1	>.1	.056	.087	>.1	.024	.024	.036	.039	.032	.077	>.1	>.1	.047	.092
SupCon $\rightarrow$ res-nplm (concat)	0.890	0.06	>.1	>.1	.063	.075	>.1	.029	.026	.027	.042	.033	.048	.093	>.1	.047	.092
SupCon+SIGReg $\rightarrow$ res (residual)	0.852	0.06	>.1	.073	.092	>.1	>.1	.025	.025	.044	.049	.032	.088	>.1	.096	.082	>.1
SupCon+SIGReg $\rightarrow$ res (concat)	0.869	0.05	>.1	.079	.072	.096	>.1	.021	.025	.036	.047	.033	.089	>.1	>.1	.075	>.1
SupCon+classwise SIGReg ($\lambda=5$)	0.832	0.05	>.1	.059	.077	>.1	>.1	.021	.023	.039	.049	.077	.096	.096	.087	.070	.073

Table 26: Galaxy10 / dino: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / DINO, 15 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.03, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	.090	.022	>.1	>.1	.150
Eucl. (anch.)	.019	.021	.030	.014	.018	.019
Maha.	>.1	>.1	.066	>.1	>.1	.150
MMD	.029	.039	.036	.059	.044	.039
SparKer	.062	.079	>.1	.048	.079	.079
SparKer (anch.)	.044	.044	.089	.035	.042	.044
<i>SupCon</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: 0.02, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	.078	>.1	>.1	>.1	.150
Eucl. (anch.)	.020	.021	.025	.020	.031	.021
Maha.	>.1	.070	.069	.085	>.1	.085
MMD	.052	.028 [†]	.066	.087	.066	.066
SparKer	.040	.083	.096	.078	>.1	.083
SparKer (anch.)	.039	.036	.049	.044	.044	.044
<i>SupCon</i> $\rightarrow$ <i>res-nplm</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.02, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	.072	.036	>.1	>.1	.150
Eucl. (anch.)	.016	.019	.038	.020	.019	.019
Maha.	>.1	>.1	.041	.068	>.1	.150
MMD	.027	.039	.040	.050	.036	.039
SparKer	.046	.084	.092	.069	.097	.084
SparKer (anch.)	.045	.037	.052	.044	.043	.044
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: never, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	.062	>.1	>.1	.150
Eucl. (anch.)	.020	.019	.022	—	.031	.021
Maha.	>.1	.075	.078	>.1	>.1	.150
MMD	.027	.029	.039	.046 [†]	.046	.039
SparKer	.048	.077	.084	.075 [†]	>.1	.077
SparKer (anch.)	.044	.046	.047	—	.052	.047
<i>SupCon+SIGReg</i> $\rightarrow$ <i>res</i> (<i>residual</i>) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	.096 [†]	>.1	>.1	.150
Eucl. (anch.)	.014	.026	.026	—	.034	.026
Maha.	.097	.086	>.1	>.1	>.1	.150
MMD	.066	.028 [†]	.047	.052 [†]	.048	.048
SparKer	.059	>.1	.074 [†]	.084 [†]	>.1	.084
SparKer (anch.)	.047	.074	.043	—	.046	.047
<i>NPLM-dist.</i> (<i>sup.</i>) — engaged from d0: never, d3: never, d5: never, d7: never, d8: 0.03; median engaged purity 0.28						
Eucl.	.071 [†]	>.1	>.1	.018 [†]	>.1	.150
Eucl. (anch.)	—	—	—	—	.030	.030
Maha.	>.1	>.1	>.1	.072 [†]	>.1	.150
MMD	.035 [†]	.024 [†]	.029 [†]	.037 [†]	.052	.035
SparKer	.045 [†]	.078 [†]	.056 [†]	.045 [†]	.094	.056
SparKer (anch.)	—	—	—	—	.047	.047

Table 27: Galaxy10 / dino: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage*: Galaxy10 / dino, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 168 (test, draw) points.

space	one label (class 2)					all seen labels				
	out. AUC	probe	Maha.	MMD	SparKer	out. AUC	probe	Maha.	MMD	SparKer
<i>lejepa</i>										
pretrained trunk	0.86	1.00	.007	.042	.022	0.89	1.00	.006	.023	.042
SupCon	0.69	0.99	.079	>.1	.036	0.51	0.95	.016	.083	>.1
SupCon+SIGReg ($\lambda=5$)	0.57	0.99	.028	>.1	.035	0.57	0.94	.031	>.1	.082
NPLM-dist. (sup.)	0.48	0.88	.013	>.1	>.1	0.58	0.86	.027	>.1	>.1
<i>visreg</i>										
pretrained trunk	0.94	1.00	.006	.047	.008	0.94	1.00	.006	.024	.016
SupCon	0.52	1.00	.022	>.1	.023	0.59	1.00	.022	.083	.056
SupCon+SIGReg ($\lambda=5$)	0.84	1.00	.007	>.1	.009	0.69	1.00	.013	>.1	.037
NPLM-dist. (sup.)	0.35	0.77	.093	>.1	>.1	0.62	0.64	.068	.086	.088
<i>dino</i>										
pretrained trunk	0.77	1.00	.010	.049	.020	0.82	1.00	.008	.027	.080
SupCon	0.61	1.00	.013	>.1	.023	0.63	1.00	.010	.076	.060
SupCon+SIGReg ($\lambda=5$)	0.68	1.00	.024	>.1	.024	0.63	0.99	.022	.059	.067
NPLM-dist. (sup.)	0.76	0.81	.031	>.1	.043	0.63	0.69	.063	>.1	.086

Table 28: Diffuse-shift detection: a systematic smear of seen-class images instead of a novel class. The smear stays almost perfectly *decodable* (probe 0.86–1.00) but the fine-tuned discovery spaces barely see it as outlying; mean-shift Mahalanobis becomes the only reliable test there, while the raw pretrained trunk detects it at $f^*=0.006$ –0.010 everywhere — the systematics monitor. SparKer prefers the semi-clustered one-label smear in every engaged comparison: the kernel tests’ class-novelty sharpness is a dependence on clusteredness. *Coverage*: Galaxy10, 24 (backbone, space, mode) cells; smear = Gaussian blur $\sigma=2$ px applied identically to ≤ 1000 seen-class test images (source rows removed from the background pool); exp-146 toy battery. *out. AUC* = min-anchor-distance ROC of smeared vs. background; *probe* = smeared-vs-clean linear AUC.

space	novel class 2 (held out)			smeared class 2 (seen, blur $\sigma=2$)			cost
	Maha.	MMD	SparKer	Maha.	MMD	SparKer	
<i>lejepa</i>							
pretrained trunk	>.1	.031	.079	.007	.042	.022	0.2×
SupCon	.029	.015	.015	.079	>.1	.036	2.4×
SupCon+SIGReg ($\lambda=5$)	.023	.015	.014	.028	>.1	.035	1.9×
NPLM-dist. (sup.)	>.1	.019	.029	.013	>.1	>.1	0.7×
<i>visreg</i>							
pretrained trunk	>.1	.034	.073	.006	.047	.008	0.2×
SupCon	>.1	.019	.031	.022	>.1	.023	1.1×
SupCon+SIGReg ($\lambda=5$)	>.1	.025	.036	.007	>.1	.009	0.3×
NPLM-dist. (sup.)	>.1	.024	.032	.093	>.1	>.1	3.9×
<i>dino</i>							
pretrained trunk	>.1	.040	.049	.010	.049	.020	0.3×
SupCon	>.1	.031	.078	.013	>.1	.023	0.4×
SupCon+SIGReg ($\lambda=5$)	>.1	.021	>.1	.024	>.1	.024	1.1×
NPLM-dist. (sup.)	>.1	.037	.054	.031	>.1	.043	0.8×

Table 29: Same label, two anomaly types: class novelty vs. diffuse smear. The reduction is per test family, not blanket: the clustering-sensitive tests lose heavily on the smear (MMD censors on 8 of 9 fine-tuned rows; SparKer costs 1.5–2.4× where both engage), while the mean-shift Mahalanobis often detects the smear *cheaper* than novelty (costs down to 0.3×) — but a mean-shift detection carries no class structure for an anchor to mean. The raw pretrained trunk sees the smear 3–5× better than it sees the same class as novelty (cost 0.2–0.3×): the systematics monitor. *Coverage*: Galaxy10, 12 (backbone, space) pairs, both anomalies built from CLASS 2: left = class 2 held out of training and injected as a novel class (exp-150 draw d0); right = class 2 seen in training, a subset smeared and injected (exp 152). The pretrained-trunk row uses the identical space on both sides; the fine-tuned rows differ in which class the fine-tune excluded (2 vs. the archived 9). *cost* = ratio of best-test f^* , smeared over novel.

backbone	probe	eucl	per-ev.	Maha.					MMD					SparKer				
				d0	d1	d2	d3	d4	d0	d1	d2	d3	d4	d0	d1	d2	d3	d4
<i>aircraft</i> — held-out classes d0=c14, d1=c43, d2=c71, d3=c96, d4=c58																		
lejepa	0.917	0.580	0.08	>.1	.023	>.1	>.1	.036	.097	.068	.047	.075	.056	.065	.036	.033	.035	.039
visreg	0.926	0.561	0.08	>.1	.021	>.1	>.1	.040	>.1	.065	.047	.073	.056	.047	.031	.034	.037	.036
dino	0.947	0.649	0.12	>.1	.029	.069	.043	.019	.082	.056	.028	.042	.036	.041	.038	.019	.026	.023
<i>cars</i> — held-out classes d0=c172, d1=c13, d2=c109, d3=c112, d4=c53																		
lejepa	0.876	0.469	0.02	.075	>.1	>.1	>.1	>.1	.028	.034	.027	.026	.027	.015	.021	.019	.015	.015
visreg	0.932	0.472	0.02	.085	>.1	>.1	>.1	>.1	.027	.028	.033	.026	.028	.019	.019	.018	.014	.019
dino	0.971	0.472	0.03	.066	>.1	.079	>.1	>.1	.038	.035	.029	.021	.021	.014	.018	.013	.012	.013

Table 30: The on-manifold corner, every point: fine-grained novelty inside the seen manifold (aircraft variants, car models). The discovery machinery is blind — min-anchor-distance ranks the novel class *below* chance on cars (eucl AUC ~ 0.47), per-event power is dead, pools starve at the natural rate — yet the class is still a dense clump: SparKer detects it on every cars draw and backbone ($f^*=0.012\text{--}0.021$) while mean-shift Mahalanobis sensors, the exact mirror of the diffuse case. Detection survives; nothing pool-able or anchor-able comes with it. *Coverage:* Pretrained ViT-B/16 trunk (identity head), single holdout, 5 draws per (dataset, backbone), 30 cells, every point shown; probe/eucl/per-ev. are means over draws. Held-out pools are tiny (aircraft ~ 33 test images/class), so toys bootstrap them — censoring here is conservative evidence.